

TOPIC

05

Chemical energetics

Questions only • Original examination pages retained

3

QUESTIONS

49

TOTAL MARKS

1

SECTIONS



TOPIC CONTENTS

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01

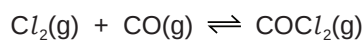
Exothermic/endothermic changes and bond energies

3 questions • 49 marks

5.1 Exothermic/endothermic changes and bond energies

#	SOURCE	MARKS	DONE	MARK	REVIEW
01	2021 Oct/Nov Paper 42 Question 4	17	☐	☐	☐
02	2023 Oct/Nov Paper 42 Question 5	16	☐	☐	☐
03	2025 Oct/Nov Paper 41 Question 4	16	☐	☐	☐

- 4 Chlorine reacts with carbon monoxide to produce phosgene gas, $\text{COCl}_2(\text{g})$. A catalyst is used.



The reaction is exothermic.

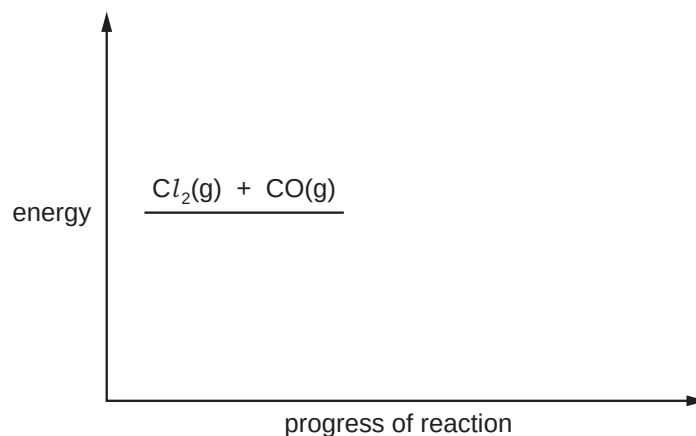
- (a) Explain why the reaction is exothermic in terms of the energy changes of bond breaking and bond making.

.....
.....
..... [3]

- (b) (i) Complete the energy level diagram for this reaction.

On your diagram show:

- the product of the reaction
- an arrow representing the energy change, labelled ΔH
- an arrow representing the activation energy, labelled A.



[3]

- (ii) State why a catalyst is used.

..... [1]

(c) Describe and explain the effect, if any, on the position of equilibrium when:

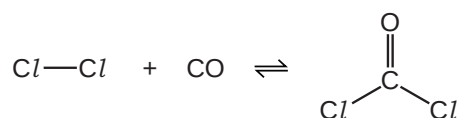
(i) the pressure is increased

.....
..... [2]

(ii) the temperature is increased.

.....
..... [2]

(d) The reaction between chlorine and carbon monoxide can be represented as shown.



When one mole of chlorine reacts with one mole of carbon monoxide, 230 kJ of energy is released.

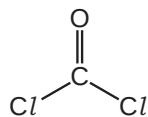
Some bond energies are shown in the table.

bond	bond energy in kJ/mol
Cl-Cl	240
C=O	745
C-Cl	400

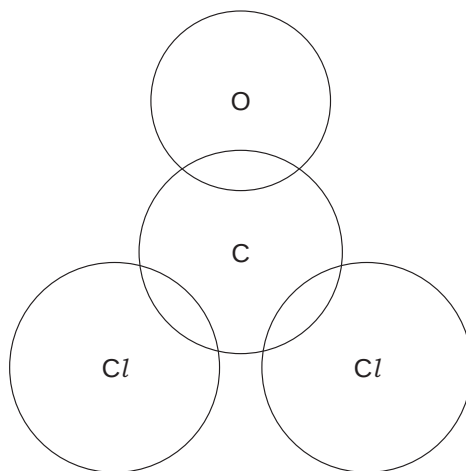
Use the information to calculate the energy of the bond between the C and the O in carbon monoxide, CO.

bond energy in carbon monoxide, CO = kJ/mol [3]

(e) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of COCl_2 .



Show outer electrons only.



[3]

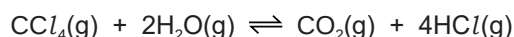
[Total: 17]

5 Chemical reactions can involve transfer of thermal energy.

(a) State the term used for the transfer of thermal energy during a reaction.

..... [1]

(b) Tetrachloromethane gas, $\text{CCl}_4(\text{g})$, reacts with steam as shown.



The reaction is reversible. The forward reaction is exothermic.

(i) State what happens, if anything, to the rate of the forward reaction if the concentration of CCl_4 is increased.

Explain your answer in terms of collision theory.

.....
.....
.....
..... [3]

(ii) State what happens to the position of equilibrium, if anything, when the pressure is increased.

Explain your answer.

.....
..... [2]

(iii) Fig. 5.1 shows an incomplete reaction pathway diagram for the forward reaction.

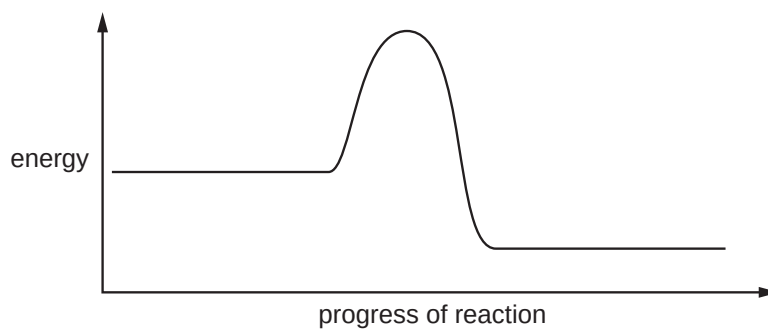
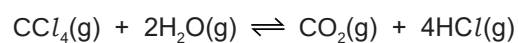


Fig. 5.1

On Fig. 5.1:

- insert the formulae of the reactants and products
- draw an arrow, labelled E_a , to show the activation energy
- draw an arrow, labelled ΔH , to show the transfer of energy in the reaction.

[3]

(iv) Define the term activation energy.

.....
..... [2]

(v) State **one** way in which the activation energy of a reaction can be changed.

..... [1]

- (c) The equation for the reaction between tetrachloromethane gas and steam can be represented as shown in Fig. 5.2.

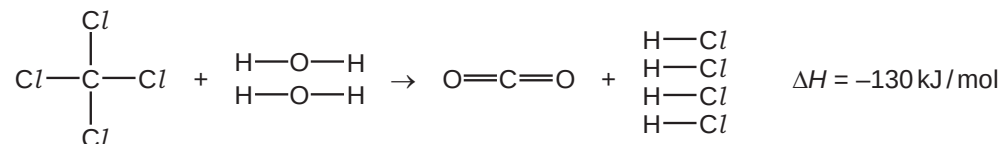


Fig. 5.2

Table 5.1 shows some bond energies.

Table 5.1

bond	C—Cl	H—O	C=O
bond energy in kJ/mol	340	460	805

Use the bond energies in Table 5.1 and the ΔH value for the reaction to calculate the H—Cl bond energy using the following steps.

- Calculate the energy needed to break the bonds in the reactants.

..... kJ

- Calculate the energy released when the bonds in carbon dioxide form.

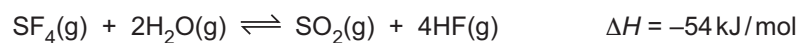
..... kJ

- Calculate the H—Cl bond energy.

..... kJ/mol
[4]

[Total: 16]

- 4 Gaseous sulfur tetrafluoride, SF_4 , reacts with steam in a reversible reaction.



- (a) Complete the reaction pathway diagram in Fig. 4.1 for this reaction.

Include in your diagram:

- the position and the formulae of the products
- an arrow, labelled E_a , to show the activation energy
- an arrow, labelled ΔH , to show the enthalpy change of the reaction.

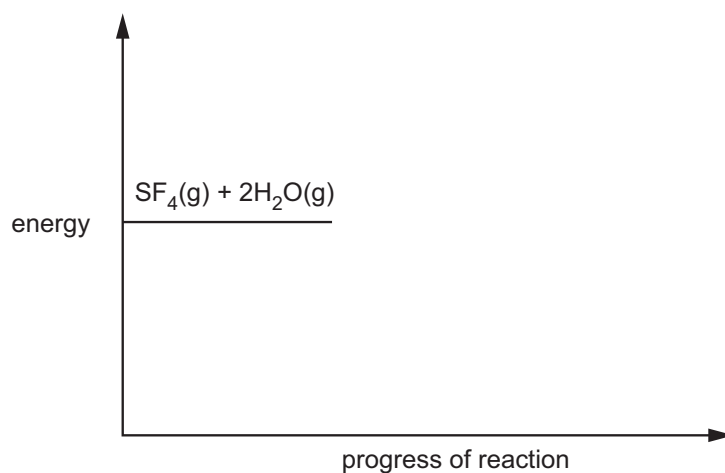


Fig. 4.1

[4]



(b) The equation for the reaction can be represented as shown in Fig. 4.2.

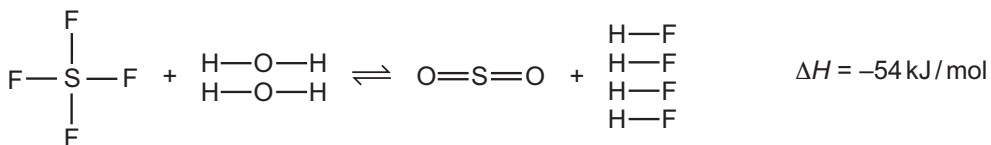


Fig. 4.2

Table 4.1 shows some bond energies.

Table 4.1

bond	S-F	O-H	H-F
bond energy in kJ/mol	330	460	570

Use the bond energies in Table 4.1 and the value of ΔH of the reaction to calculate the S=O bond energy in kJ/mol.

Use the following steps.

- Calculate the energy needed to break the bonds in the reactants.

..... kJ

- Calculate the energy released when the bonds in the products form.

..... kJ

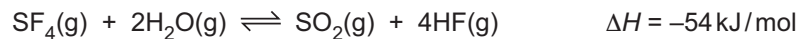
- Calculate the S=O bond energy.

..... kJ/mol
[4]





(c) The equation for the reaction is shown.



State the effect, if any, on the position of equilibrium when the following changes are made.

Give a reason for each of your answers.

- The temperature is increased.

.....

.....

- The pressure is increased.

.....

.....

- A catalyst is added.

.....

.....

[5]

(d) Explain, in terms of collision theory, why reducing the temperature decreases the rate of the reverse reaction.

.....

.....

.....

.....

.....

[3]

[Total: 16]



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Question 5 starts on the next page.

